
Adverse Event Response Standard Operating Procedure

SOP ID: SOP-AE-001

Revision: A

Effective: 2026-06-05

Owner: site principal investigator

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Abstract.

This SOP template defines the response to adverse and serious adverse events in a Physical AI oncology trial: who acts, in what order, and within what time.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 Purpose and scope

This procedure governs the detection, response, and reporting of adverse events during gated robot-assisted procedures. It applies to every site role from the moment an event is detected.

2 Roles and responsibilities

Three roles share a single response.

Lead directs the response and decides escalation

ASSIST executes the immediate safe-state action

Reserve documents the event and prepares the report

3 Procedure

The response is a timed sequence.

1. **Detect.** the gate or a clinician flags the event and the robot enters a safe state
2. **Stabilize.** the Assist role executes the immediate action within the contact-force budget
3. **Escalate.** the Lead returns the interaction to a qualified human if the gate ESCALATES
4. **Report.** the Reserve role files the adverse-event record within the required window

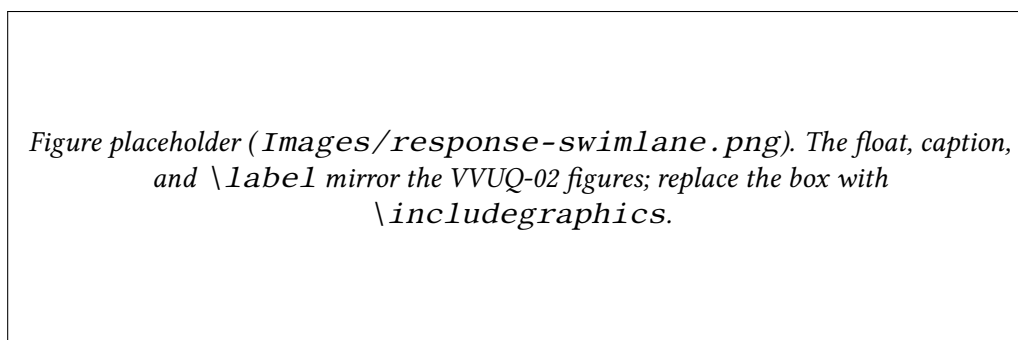


Figure 1: Figure 1. Placeholder swimlane of the timed adverse-event response.

4 Escalation and reporting

Grade	Action	Timing
mild	document and monitor	same session
serious	safe state; escalate to human	immediate
catastrophe	hard BLOCK before execution	pre-execution
predicate		

Table 1: Table 1. Grade, action, and timing (modeled on VVUQ-05 Table 1).

5 Records

Every event and response is recorded in a hash-chained audit trail; records are synthetic or illustrative in this template and carry no patient-identifiable information.

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Ethical Disclosures

The author declares no competing interests. This is an independent draft and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. All supporting numbers are simulation or illustrative results.

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Cite This Procedure

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Data Availability

Supporting records are openly available in the `kevinkawchak/cancer-automated` and `kevinkawchak/physical-ai-oncology-trials` GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

- [1] K. Kawchak, “VVUQ Processing Priority over Code Generation: Method and Pipeline.”
<https://doi.org/10.5281/zenodo.20372501>, 2026.
<https://doi.org/10.5281/zenodo.20372501>.
- [2] K. Kawchak, “Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ.”
<https://doi.org/10.5281/zenodo.20421754>, 2026.
<https://doi.org/10.5281/zenodo.20421754>.
- [3] K. Kawchak, “Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H. R. 9510).” <https://doi.org/10.5281/zenodo.20535429>, 2026.
<https://doi.org/10.5281/zenodo.20535429>.
- [4] K. Kawchak, “National Physical AI Oncology Trial Platform.”
<https://doi.org/10.5281/zenodo.19244918>, 2026.
<https://doi.org/10.5281/zenodo.19244918>.
- [5] Creative Commons, “Attribution 4.0 International (CC BY 4.0).”
<https://creativecommons.org/licenses/by/4.0/>, 2013.
<https://creativecommons.org/licenses/by/4.0/>.
- [6] L. Lamport, *LaTeX: A Document Preparation System*. Addison-Wesley, 2nd ed., 1994.